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Soundscape Overview and Glider Data

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Some Useful Links

PAMGuide:

<https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210X.12330>

YAWN:

<https://github.com/PIFSC-Protected-Species-Division/SPACIOUS-NoiseProcessing/tree/main>

Burrard Inlet Reports:

<https://twnsacredtrust.ca/wp-content/uploads/2023/06/2022-04-21-Burrard-Inlet-underwater-noise-monitoring-final-report.pdf>

JASCO Pocket Acoustics Guide:

<https://www.oceansinitiative.org/wp-content/uploads/2012/07/PocketBook-3rd-ed.pdf>

Marine Mammal Acoustics in a Noisy Ocean

Marine Mammals in Noise Book: <https://link.springer.com/book/10.1007/978-3-031-77022-7>



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Goals

01

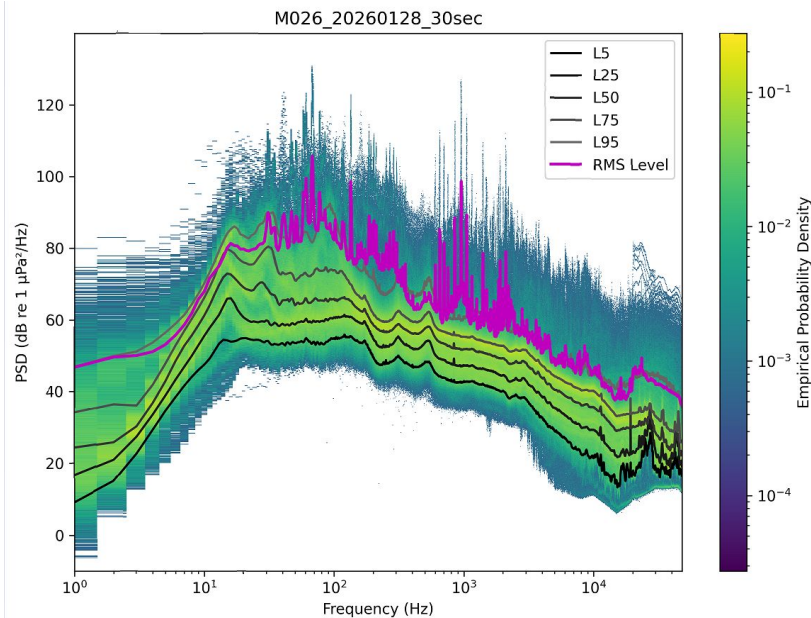
Understand how sound data are collected and build intuition

02

Get an idea of different tools available

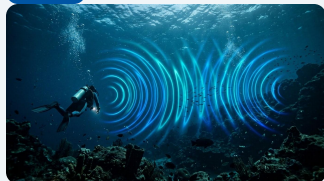
03

Understand Glider Rodeo Data structure



Underwater Sound Processing Pipeline

01



Underwater Sound

Varying Pressure (Pa)

Acoustic waves travel through water causing subtle continuous fluctuations in hydrostatic pressure.

02

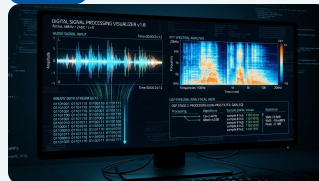


Hydrophone

Voltage Signal (V/Pa)

Piezoelectric elements convert physical pressure oscillations into continuous electrical voltage signals.

03

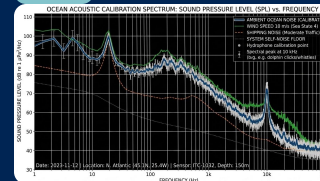


ADC / Storage

Digital Bits (bit/V)

Analog-to-digital converter samples voltage, quantizing it into digital binary bits stored on computer files.

04



Calibration

Calibrated SPL (Pa/Bit)

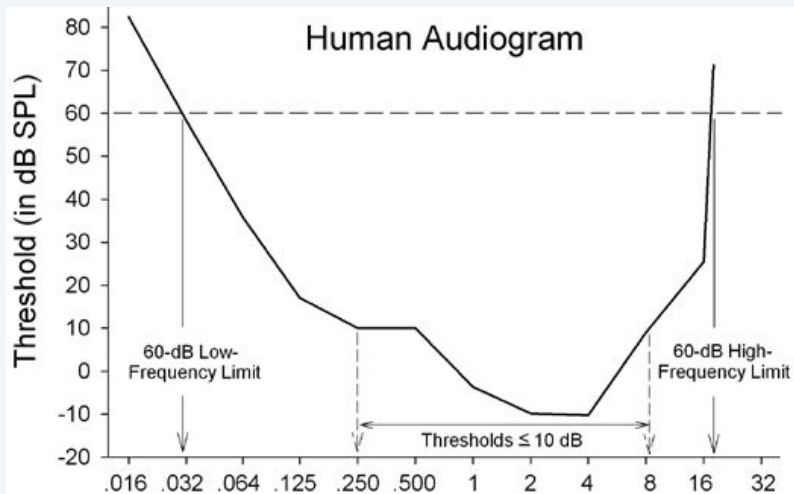
System sensitivity curves apply scaling to digital bits, recovering true Ocean Sound Pressure Level (dB re 1 μ Pa).



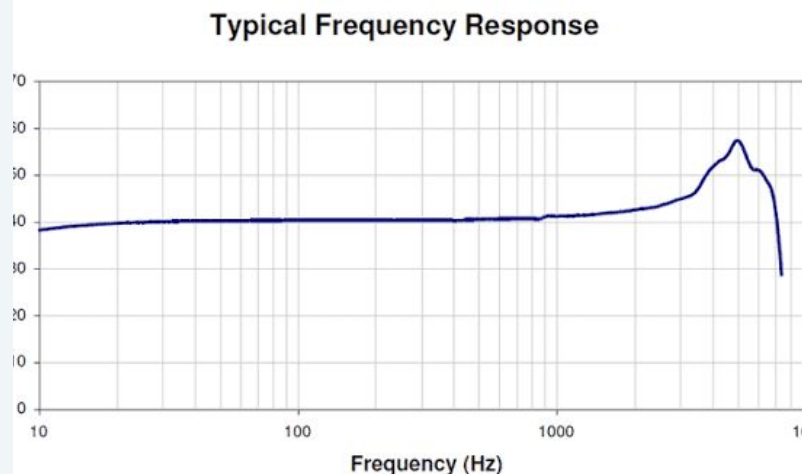
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Hydrophone Sensitivity & Calibration

Human Audiogram (Ear)



Typical Frequency Response (Hydrophone)



Building Intuition

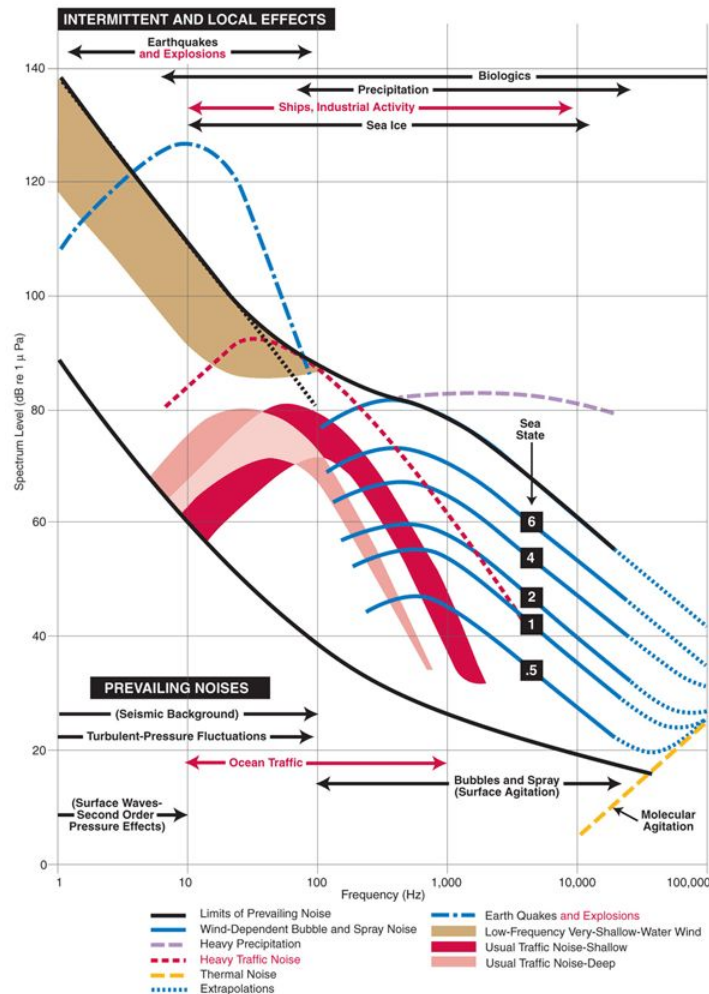
01 Measurement Diversity
Dozens of ways to measure sound (see JASCO pocket guide)

02 Sound Types
Impulsive and continuous sound classifications

03 Standard Units
Generally Spectrum Level (SPL dB/Hz), Power Spectrum Density (PSD dB/ $\sqrt{\text{Hz}}$)

04 Ambient noise (generally) decreases with increasing frequency

05 Key Threshold Warning
120 PSD or SPL levels regularly > 130 dB should raise alarm bells



Yet Another Way to analyze Noise (YAWN)

Merchant et al. 2015

- Creates PSD, SPL, TOL, waveform
- Stats plots: mean, percentiles, SPD, CDF
- Writes to csv
- Single calibration value

Updated 2020

- GUI for reproducibility
- Saved to mat files
- Added option to save to HMD
- Fixed bug in TOL
- Created various plots
- Cross validated against original outputs

Burrard Inlet Underwater Noise
Characterization: 2021 Final Report
Prepared for the Vancouver Fraser Port Authority and
the Tsleil-Waututh Nation

SMRU Consulting North America



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What YAWN Is and Isn't



Is

- A Python package capable of producing standard metrics and producing plots (HMD, SPD, $\frac{1}{3}$, decade, octave, broadband)
- Flexible (dates, duty-cycles, calibration)
- Simple (ish)
- Gs bucket enabled
- Produces widely sharable HDF5 files (parent of NetCDF)



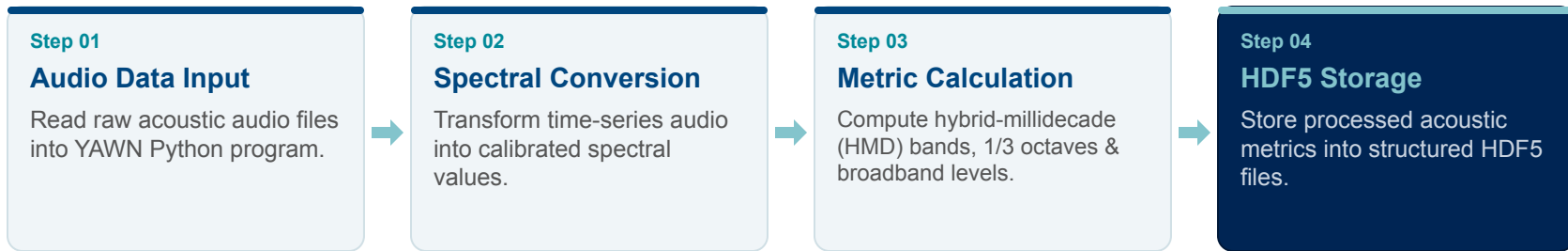
Isn't

- Well supported
- Able to automatically retrieve calibration values
- Perfectly aligned with PyPam and other tools (none are)
- 1/3rd octaves not ANSI standard

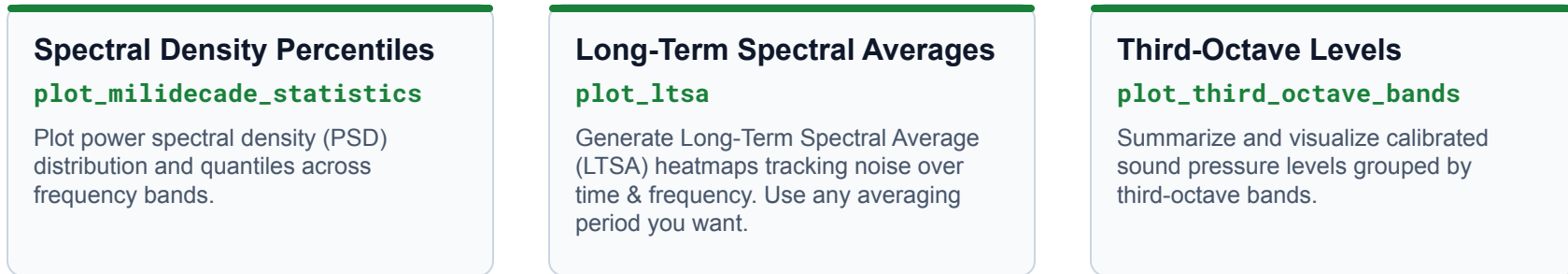
Yet Another Way to Analyze Noise (YAWN)

<https://github.com/PIFSC-Protected-Species-Division/SPACIOUS-NoiseProcessing>

1. Data Processing & Storage Pipeline



2. YAWN Plotting & Visualization Functions



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YAWN HDF5 File Structure

HDF5 = Hierarchical Data Format version 5



Project Level (sg607_20260128_5min)

Deployment Group (GliderRodeo)

Attributes / Metadata

Parameters used to run YAWN:

- Audio directory / source
- Output location
- Calibration settings
- Project & deployment metadata

Datasets (Array Data)

- **DateTime**
- **broadband**
- **decadeLevels**
- ***decadeFreqHz***
- **hybridMiliDecLevels**
- ***hybridDecFreqHz***
- **thirdoct**
- ***thirdOctFreqHz***



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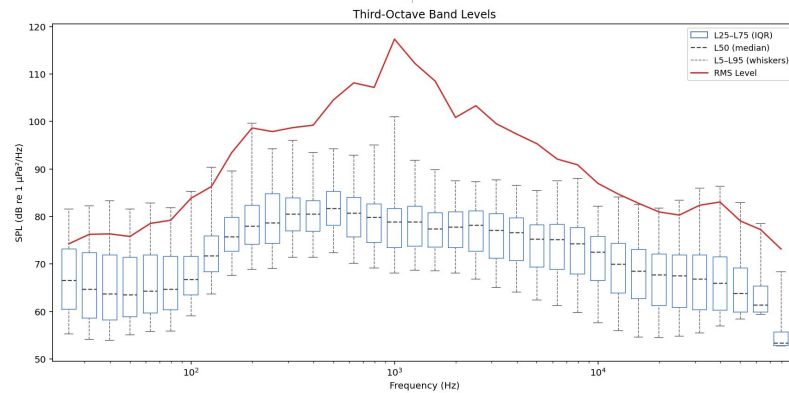
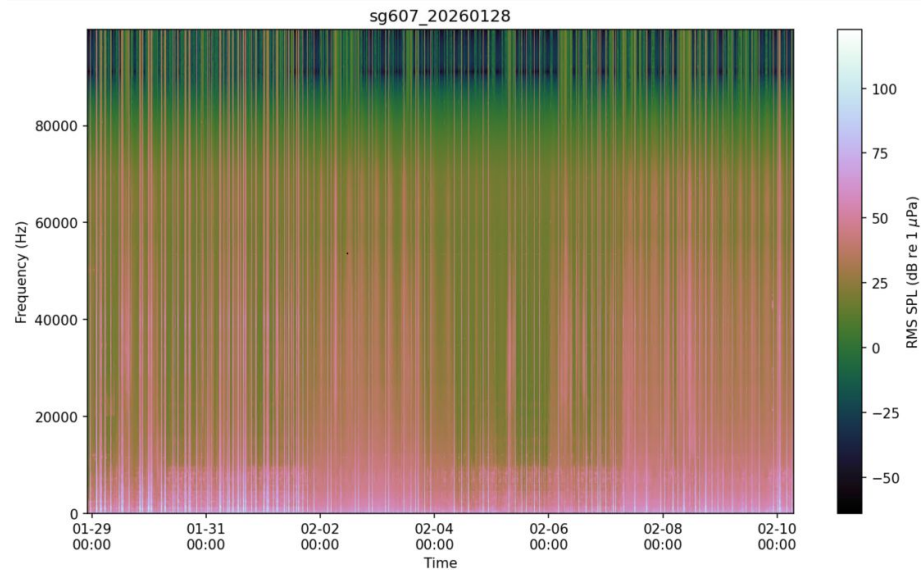
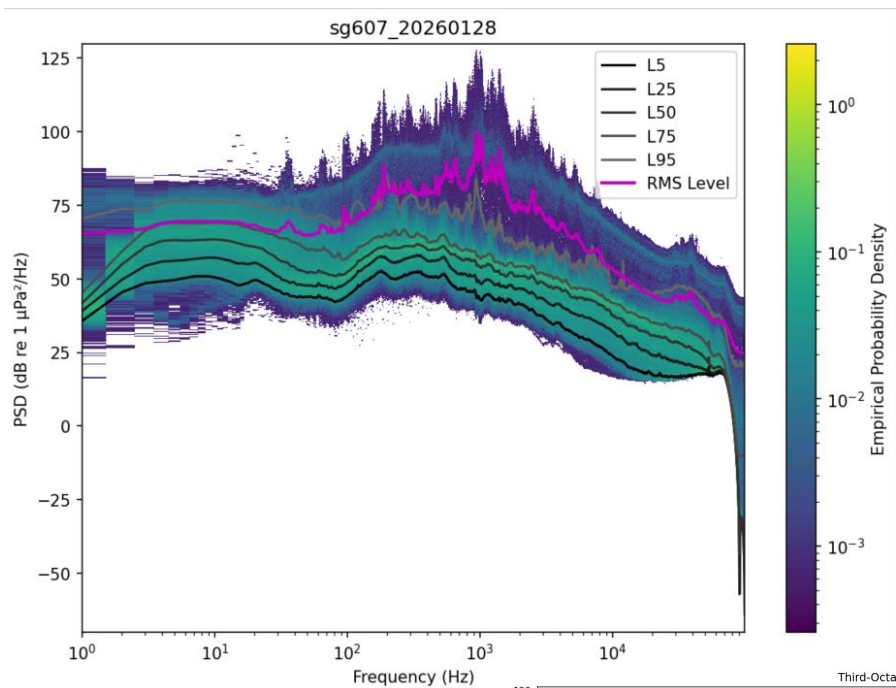
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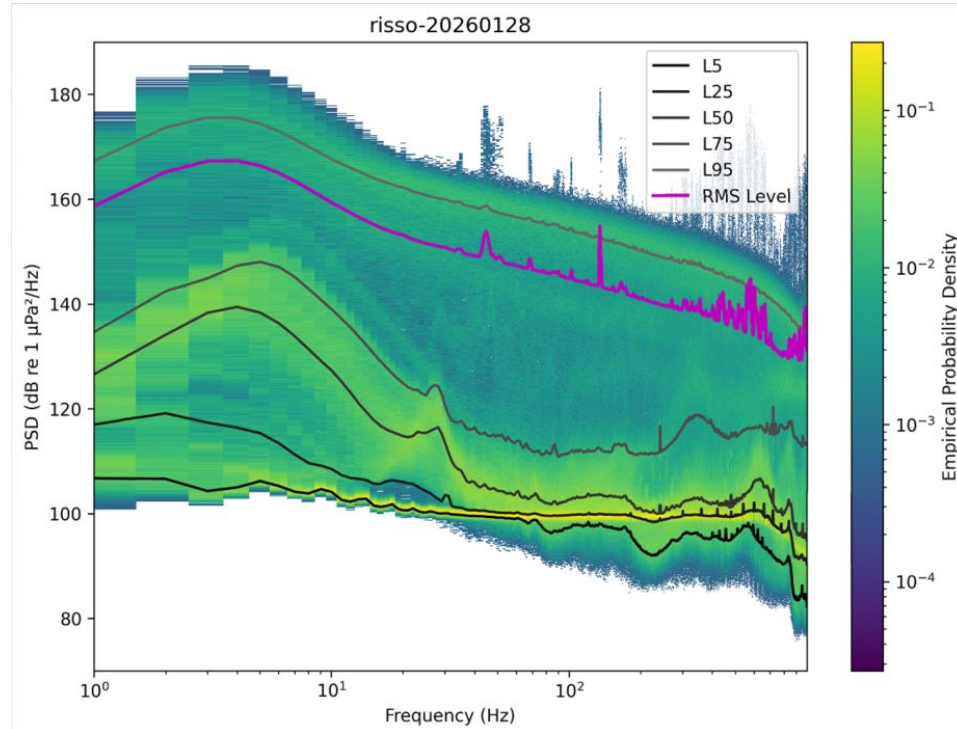
- `summarize_hdf5_file`: Summarize file structure
- `List_hdf5_deployments`: List the deployments in the file
- `plot_milidecade_statistics`: look at PSD distribution over frequency
- `plot_third_octave_bands`: summarize third-octave levels by band
- `plot_ltsa`: see noise over time and frequency as a heatmap
- `export_metric_csv`: Export a given metric to csv file



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Example Setup

```
from noiseProcessGoogleCloud import NoiseApp

# Define the location of the data on google cloud
gsCloudLoc = "gs://swfsc-1/2024_CalCurCEAS/glider/audio_flac/sg680_CalCurCEAS_Sep2024"

# Define where you want to store the data
out_dir = r"C:\Users\pam_user\Documents\HybridMilliDaily"

# Location of the calibration csv, first column should be frequency in Hz
# Second column should be end-to-end calibration in dB
calib_csv = 'C:\Users\pam_user\Downloads\sg680_CalCurCEAS_Sep2024_sensitivity.csv'

# Declare the noise app object and give it a project name and
# a deployment name. Note that you can store multiple
# deployments within a project
app = NoiseApp(
    Si=calib_csv,
    soundFilePath=gsCloudLoc,
    ProjName='sg680_CalCurCEAS_Apr2022',
    DepName='SG680',
    DatabaseLoc=out_dir,
    rmDC=True, # Remove the DC offset from each audio file
    Si_units='V/μPa'
)

# Go do the thing!
app.run_analysis()
```



Other Useful Tools for Noise Analysis

PyPAM (Well supported, python)

Triton (Scripps, matlab and standalone)

PAMGuard (SMRU, Java, Industry standard, well supported by two blokes in Scotland. Breaks often)

RAVEN (Cornell, Java, not open source fantastic for viewing spectrograms, possibly other)

MANTA (hum.... Not sure what happened to it)